

Viva / Q&A Preparation — Likely Questions with Answers

Project: Cross-Dataset Generalization of ML-Based Intrusion Detection Systems **Use:** Practise these out loud. Answers are written in a natural speaking style.

A. Project basics (almost certain)

Q1. What is your project about? > We study why machine-learning intrusion detection systems fail when moved from one network to > another, and we use bio-inspired optimisation (genetic algorithms, particle swarm) to choose > the features that let a model work across different networks instead of just one.

Q2. What is an intrusion detection system? > A system that monitors network traffic and raises an alarm when it sees malicious activity. A > network IDS (NIDS) looks at flows of traffic and classifies each as benign or an attack.

Q3. Why machine learning instead of traditional signature-based detection? > Signature-based systems only catch attacks whose “signature” is already known. ML can learn > patterns from examples and can flag novel or slightly modified attacks. But ML has its own > weakness — poor generalisation — which is exactly what we study.

Q4. What problem are you solving? > The generalisation gap. Models get 99% when tested on the same dataset but often below 40% — > sometimes near random — on a different dataset. We want to measure this gap and reduce it.

Q5. What is your main contribution / novelty? > Metaheuristic feature selection for NIDS has always been optimised for *same-dataset* accuracy. > We are the first to use a **cross-dataset fitness function** — we select features that transfer > to unseen networks. We also build a common feature subspace across three very different > datasets and characterise what transfers.

B. Fundamental concepts

Q6. What is a network flow? > A group of packets that belong to the same conversation, identified by source/destination > IP and port and protocol. Instead of analysing every packet, we summarise the flow with > statistics (duration, bytes, packets, inter-arrival times).

Q7. What is a feature? Give examples from your datasets. > A numeric column describing a flow. CIC-IDS-2017 has 78 (e.g. Flow Duration, > Total Fwd Packets); UNSW-NB15 has 42 (e.g. dur, sbytes); TON_IoT has 42 Zeek fields > (e.g. duration, src_bytes, conn_state).

Q8. What is supervised learning here? > We train a model on labelled flows (benign vs attack) so it learns to predict the label of a > new flow. We use classifiers like Random Forest, XGBoost, SVM and a small neural network.

Q9. What is overfitting, and why does it matter here? > Overfitting is when a model memorises training data instead of learning general patterns. On a new network it fails. Dataset-specific features make overfitting worse — a key reason for the generalisation gap.

C. Cross-dataset generalisation

Q10. Explain cross-dataset generalisation with an example. > Train a model on UNSW-NB15 (an Australian lab network) and test it on TON_IoT (an IoT testbed). > The model has never seen TON_IoT's traffic patterns; if it learned real attack behaviour it will still work — if it learned UNSW-specific shortcuts, it will fail.

Q11. Why do models fail cross-dataset? > Three reasons: (1) **different feature spaces** — the columns don't even match; (2) **dataset-specific shortcuts** — IPs, ports and collection-window features that don't exist or don't mean the same thing in another network; (3) **different attack distributions** — the mix of attacks and benign traffic differs.

Q12. Is it even possible to fully solve this? > Probably not fully — the distributions genuinely differ. Our goal is to *measure* the gap and to *narrow* it using transfer-oriented feature selection, and to report honest, relative improvements over baselines rather than claiming perfect accuracy.

Q13. What is domain adaptation, and why not just use that? > Domain adaptation aligns feature distributions between a source and target dataset. It works, but usually needs some target-domain data and complex neural training, and assumes label semantics overlap. We chose feature selection because it is simpler, interpretable, and directly identifies *which features* are the problem — which also gives practical guidelines.

D. Datasets

Q14. Which datasets and why? > CIC-IDS-2017, UNSW-NB15 and TON_IoT. They are the most-used modern benchmarks and are **structurally different** (different networks and different feature-extraction tools: CICFlowMeter, Argus/Bro, and Zeek), which is essential for studying generalisation.

Q15. How big are they? > CIC-IDS-2017: 2,830,743 flows, 78 features (one duplicated), 15 classes. UNSW-NB15: 257,673 records, 42 features, 9 attack classes plus Normal. TON_IoT: 211,043 records, 42 features, 10 classes.

Q16. What problems did you find in the data? > CIC-IDS-2017 has 4,376 infinite and 1,358 missing values (division by zero in rate columns), a duplicated Fwd Header Length column, a non-ASCII label character, and very severe class imbalance (only 11 Heartbleed and 36 Infiltration samples). TON_IoT is nearly balanced (20,000 per attack). None have missing values apart from CIC-IDS-2017.

Q17. Why is the class imbalance a problem? > If 80% of CIC-IDS-2017 is benign, a lazy model can get 80% accuracy by always saying > “benign”. That’s why we use balanced accuracy, macro-F1 and per-class F1 instead of plain > accuracy.

Q18. What is the “feature-schema mismatch”? > The three datasets share **no common column name**. CICFlowMeter computes statistics; Argus/Bro > computes counters; Zeek logs raw protocol fields. Only about 6 raw features are semantically > equivalent across all three. This mismatch is the root cause of cross-dataset failure.

E. Methodology, feature selection, metaheuristics

Q19. What is feature selection and why use it? > Choosing a smaller, more useful subset of features. It makes models faster and less prone to > overfitting. For us it is the key lever: if we keep only features that are meaningful on *any* > network, the model transfers better.

Q20. What is a metaheuristic? Why not brute force? > A nature-inspired search that finds good solutions without checking all possibilities. With 20 > features there are over a million subsets, so brute force is wasteful; GA and PSO find good > subsets quickly.

Q21. How does a Genetic Algorithm work? > It keeps a population of candidate feature-subsets. Each is scored (fitness). The best ones > “reproduce” via crossover (mixing two subsets), and we add random mutations, over many > generations. Good subsets survive and improve.

Q22. How does PSO work? > It simulates a flock of birds. Each “particle” is a candidate subset moving through the search > space, and each is pulled toward its own best position and the swarm’s best. It converges on > good regions of the search space.

Q23. What exactly is your fitness function? > It is the average cross-dataset performance: we train a fast model (Random Forest or XGBoost) > on the source dataset using only the selected features, test it on the target dataset, and > use balanced accuracy / macro-F1. The optimiser is rewarded when the subset transfers well. > This is the crucial difference from previous work, which uses same-dataset accuracy.

Q24. Why standardise features per dataset and not globally? > To avoid leaking target-dataset information into training. Each dataset is normalised using its > own statistics.

Q25. What is the “shared subspace”? > A common set of semantically equivalent features — duration, source/destination bytes, source/ > destination packets, protocol, service, connection state, plus derived ratios — about 15–25 > dimensions after encoding. We run cross-dataset experiments in this common space.

Q26. What are “leakage” or “shortcut” features? > Features that accidentally reveal the answer without real detection — e.g. IP addresses, port > numbers, timestamps, or collection-window counters. A model can “cheat” with them on one > dataset but fail elsewhere. We remove them.

Q27. Why remove the destination port if it's a real network feature? > Ports are highly dataset-specific (a lab's attack used a particular port). Keeping them lets > the model memorise the scenario instead of behaviour. Research [3] showed models relying on > port categories that were 96–435× more common in the source attacks — a classic shortcut.

F. Evaluation

Q28. What models will you use and why? > Random Forest, XGBoost/LightGBM, SVM, Logistic Regression and a small MLP. Classic models are > fast enough to sit inside a GA/PSO fitness loop and are easy to interpret; the MLP gives a > neural baseline.

Q29. What metrics and why not accuracy? > Balanced accuracy, macro-F1, and per-class F1. Because the datasets have very different and > severe class imbalances, plain accuracy hides failures on rare classes.

Q30. How will you test cross-dataset? > Train on one dataset, test on another, for all ordered pairs — a 3×3 matrix (the diagonal is > same-dataset). We repeat with fixed seeds and report mean and standard deviation.

Q31. What are your baselines? > Feature selection baselines: no selection, mutual information, random-forest importance, PCA, > RFE and LASSO — each matched to the same subset size. The comparison tells us whether > transfer-optimised selection actually beats standard selection.

Q32. What does “success” look like? > Reproducing the documented collapse, then showing a **statistically reliable improvement** in > cross-dataset F1 for the metaheuristic-selected features over (i) no feature selection and (ii) > the best standard feature-selection baseline. Also, identifying which features/attacks transfer.

G. Literature-specific questions (they can ask about ANY paper)

Q33. What did Cantone et al. (2024) find? [1] > They benchmarked four classifiers across four datasets (CIC-IDS-2017, CSE-CIC-IDS2018, > LycoS-IDS2017, LycoS-Unicas-IDS2018) using a restricted common feature set. Same-dataset > results were near-perfect; cross-dataset results fell to near random chance. They concluded > the cause is dataset-specific artefacts, and they only *diagnosed* the problem.

Q34. How is Cantone different from you? > Same problem, but they use a fixed common feature set and do nothing to improve transfer. > We add an optimiser that *selects features for transfer*, and a third dataset (TON_IoT).

Q35. What did Hossain et al. (2026) do? [2] > They evaluated Random Forest, Logistic Regression and Naïve Bayes on UNSW-NB15 and TON_IoT. > RF got 95.08% (UNSW) and 99.79% (TON) in-dataset but under 40% cross-dataset. Pure > measurement, no mitigation.

Q36. What did Hakim et al. (2026) show? [3] > On three IIoT datasets, lightweight models relied heavily on coarse port-category features — > a shortcut — and the evaluation protocol (how imbalance is handled) could even reverse which > target looked harder. Adversarial robustness was unrelated to cross-network generalisation.

Q37. What did Butt et al. (2026) find? [4] > Evaluated tabular representation learning (including TabICL), autoencoders and transformers for > NetFlow intrusion detection. Transfer sometimes worked, but depended strongly on the specific > source→target pair — no universal winner.

Q38. What did Fatima & Ali (2022) do? [8] > Used a wrapper Genetic Algorithm to select features with stacking/bagging ensembles on > UNSW-NB15 and CICDDoS2019, improving multi-class accuracy and detection rate — but the fitness > was same-dataset accuracy.

Q39. What is NSGA-II used for in the literature? [9] > Suman et al. (2019) used NSGA-II to optimise three information-theoretic objectives at once for > unsupervised feature selection on KDD-99/NLS-KDD/Kyoto, obtaining Pareto-optimal subsets. It > motivates our optional multi-objective version (F1 vs number of features).

Q40. What is MPDGGA? [10] > A multi-population, diversity-guided Genetic Algorithm by Li (2026) that maintains diversity to > select very few features while getting the best accuracy on 10 of 11 datasets — all > single-dataset.

Q41. What did Yang et al. (2026) optimise? [12] > NSGA-III jointly optimised a LightGBM feature threshold and hyper-parameters for weighted F1, > inference latency and model size on EV-charging and CICIDS2017 data — efficiency objectives, > not generalisation.

Q42. What is DI-NIDS? [13] > Layeghy et al. use adversarial domain adaptation to learn domain-invariant features, then a > One-Class SVM to detect anomalies, improving cross-domain performance — but it needs aligned > feature spaces and target-domain data.

Q43. Why cite Ring et al. (2019)? [14] > Their survey of NIDS datasets explains that datasets differ in collection environment, feature > semantics and labelling — the fundamental reason cross-dataset generalisation is hard.

Q44. What did Vitorino et al. (2024) show? [15] > Using a corrected CICIDS2017 (“NewCICIDS”) changed model rankings compared to the original, > showing the dataset’s flaws materially affect conclusions — which justifies our careful > cleaning and leakage removal.

G2. Our own results (be ready to defend these)

Q. What did your baseline show? > We trained five ensemble models (Random Forest, Extra Trees, Histogram Gradient Boosting, > XGBoost, soft-voting) on the **full** datasets. Same-dataset accuracy was **0.96–0.97**, but > cross-dataset was only **0.37–0.42**, with balanced accuracy **0.46–0.49 — essentially random.** > XGBoost trained on CIC-IDS-2017 and tested on UNSW-NB15 had balanced accuracy exactly > **0.5000**. This reproduces Cantone et al. [1] and Hossain et al. [2].

Q. What is your metaheuristic approach, and what did it achieve? > Our method is **DA-MFS (Domain-Aligned Metaheuristic Feature Selection)**. First we align each > dataset’s feature distributions (a per-dataset quantile/rank transform to a standard normal) to > remove distribution shift. Then a Genetic Algorithm (and a binary PSO) search for a feature > subset that maximises a **cross-dataset fitness** (mean of balanced accuracy and macro-F1 over > the six train→test pairs). The GA selected a **4-feature** subset and, on the full data, > improved cross-dataset **accuracy from 0.373 to 0.508 (+13.6 pp)**, **balanced accuracy by +10.4 pp**, **macro-F1 by +15.3 pp**, and **attack recall by +12.8 pp** over the raw baseline.

Q. Why did plain feature selection only give a small gain? > Because features from different datasets have different *marginal distributions* even when they > are semantically the same. Selecting features does not fix that. Alignment does: alignment alone > raised accuracy from 0.373 to 0.478; the metaheuristic added a further +3.1 pp accuracy and > +7.6 pp macro-F1 while shrinking the representation to 4 features.

Q. Is DA-MFS novel? > Yes. We found **no prior work** combining quantile-based domain alignment with metaheuristic > feature selection under a cross-dataset fitness for NIDS (arXiv queries for “domain adaptation + > feature selection + intrusion detection” and “domain-invariant + feature selection + > genetic/evolutionary” return no results). Quantile/rank transport is used in tabular domain > adaptation (KRDA), but not with evolutionary feature selection for IDS.

Q. Why optimise on a sample and evaluate on the full data? > Each fitness evaluation trains a model; doing that thousands of times on 2.83M rows is > intractable. We optimise on a stratified **20,000-row sample per dataset** and validate the > winning subset on the **full** datasets — the full-data numbers are what we report.

Q. Why does attack recall now improve? > The DA-MFS fitness averages balanced accuracy and macro-F1, both of which reward recall, and the > aligned features remove the distribution artefacts that made recall unstable. Recall rises from > 0.479 to 0.607.

Q. Did GA beat PSO? > On the aligned sample, GA reached fitness **0.551** vs PSO **0.545**, so we used the GA subset. > A fuller comparison across seeds and objectives is part of the end-semester work.

H. Tough questions & how to handle them

Q45. If cross-dataset accuracy stays near random, is your project a failure? > No. Our success metric is *relative improvement over baselines*, not absolute accuracy. > Characterising the gap, identifying transferable features, and producing guidelines are > valuable contributions even if absolute numbers remain modest.

Q46. Could deep learning solve it better than feature selection? > Deep models are also affected — they overfit the same shortcuts. Feature selection is > complementary and interpretable, and it directly targets the identified cause. We include an > MLP to compare, and note deep approach as future work.

Q47. How do you know you removed *all* leakage? > We remove obvious identity features (IPs, ports, IDs, timestamps) and use SHAP/importance to > check for remaining suspiciously

dominant features. We also run an ablation with and without > leakage features to quantify the effect.

Q48. What about the different attack labels across datasets? > We first map everything to binary (benign vs attack). For multi-class we use a coarse taxonomy > (DoS/DDoS, Recon/Scanning, Brute-force, Backdoor, Web/Injection, etc.). Some categories don't > map cleanly — we report that as a finding.

Q49. Why is your shared subspace so small? > Because the datasets use different tools. Only fields like duration, byte/packet counts and > protocol/service/state are truly comparable. A small, honest subspace is better than forcing > incompatible features together.

Q50. What if the optimiser just overfits the target dataset? > We design the protocol so the target is only used for *scoring*, and we use a held-out source > fold for training. We report results across multiple seeds and target pairs, not a single > lucky split.

Q51. How is this different from just combining all datasets into one big training set? > Combining still leaves mismatched columns and different scales; you'd need alignment anyway. > Our shared-subspace approach makes the alignment explicit and tests transfer *without* > target labels — a stricter, more realistic setting.

Q52. Which part did each team member do? (*prepare a real division of work*) > e.g. Dataset acquisition/cleaning and analysis; literature review and methodology design; > implementation of models and optimisers. Be ready to name specific files/scripts each person > wrote.

I. Logistics

Q53. What is your timeline? > Cleaning and harmonisation (weeks 3–6), cross-dataset baseline (weeks 6–8), metaheuristic > feature selection (weeks 8–11), baseline comparison and analysis (weeks 11–14), final report > and guidelines.

Q54. What tools are you using? > Python with pandas/NumPy/scikit-learn/XGBoost/Light-GBM for data and models, DEAP for the > Genetic Algorithm and PySwarms (or custom code) for PSO, and Matplotlib/Seaborn for figures. > Everything is in a version-controlled repository.

Q55. What have you completed so far? > Dataset analysis (real measured statistics and charts), the 15-paper literature review with the > identified gap, the objectives and 7-phase methodology, and the reproducible codebase.

J. One-line memory hooks

- **Problem:** 99% in the lab, ~random in the real world.
- **Cause:** mismatched features + dataset-specific shortcuts (leakage).
- **Solution:** select a shared set of transferable features.
- **How:** GA / PSO with a **cross-dataset fitness function**.

- **Novelty:** nobody has optimised features *for transfer* in NIDS before.
- **Success:** beat the baselines and explain what transfers.